



Kuwait University
Faculty of Science
Department of Mathematics

Math 101: Calculus I

Summer 2021/2022

First Exam
Monday, July 4, 2022

Name										
Student ID										
Section No.						Serial No.				

Duration: 90 minutes.

Give full justification for your answers and state clearly any theorems you use.
Calculators and communication devices are not allowed in the examination room.

Question 1	
Question 2	
Question 3	
Question 4	
Question 5	
Question 6	
Question 7	
Question 8	
Total	100

1. [10 pts.] Simplify the expression $\tan(\sin^{-1} x)$.

2. [10 pts.] Let $f(x) = \frac{2^x - 3}{2^x - 1}$. Find $f^{-1}(x)$.

3. [10 pts.] Use the Intermediate Value Theorem to show that the equation $x^3 + x^2 - 2x - 1 = 0$ has a real root.

4. [10 pts.] Find a value for k , if any, such that $\lim_{x \rightarrow 1} \left(\frac{1}{x-1} - \frac{k}{x^2-1} \right)$ exists and is finite.

5. [3+7=10 pts.] Let $f(x) = \frac{|x|}{x^2 - x}$.

(a) Find all points of discontinuity of f , if any.

(b) Classify each discontinuity as removable, jump, or infinite (justify your answers).

6. [8+7 = 15 pts.] Evaluate each of the following limits, if it exists.

(a) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x + 2} - 2}.$

(b) $\lim_{x \rightarrow 0} x^2 e^{\sin(\frac{\pi}{x})}.$

7. [7+8 = 15 pts.] Let $f(x) = \frac{\sqrt{4x^2 + 10}}{x - 5}$.

(a) State the domain of f , and find the vertical asymptotes of the graph of f , if any.

(b) Find the horizontal asymptotes of the graph of f , if any.

8. [10+10 = 20 pts.] Use the graph of $y = f(x)$ shown below to answer the followings:

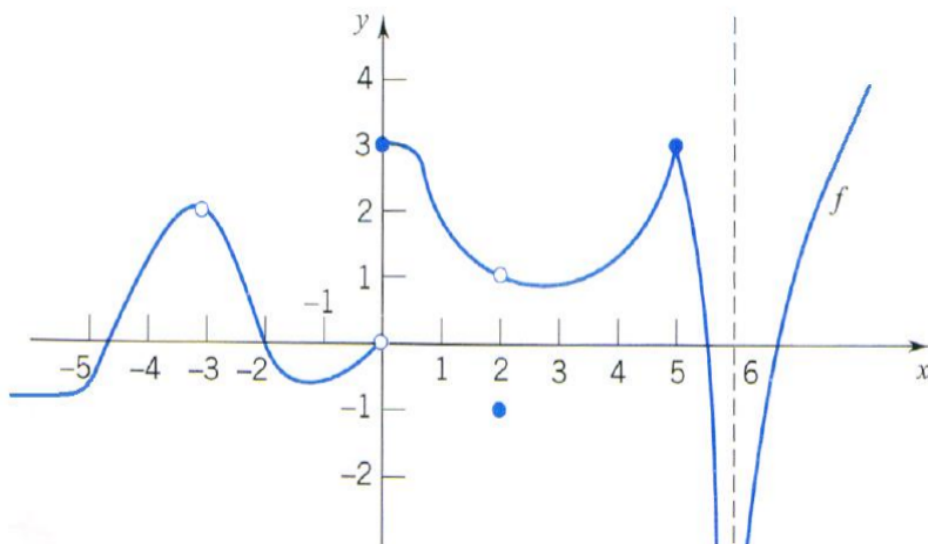


Figure 1: The graph of $y = f(x)$.

a) Find each of the following limits, if it exists:

i) $\lim_{x \rightarrow -3} f(x) =$

ii) $\lim_{x \rightarrow 0^-} f(x) =$

iii) $\lim_{x \rightarrow 0^+} f(x) =$

iv) $\lim_{x \rightarrow 2} f(x) =$

v) $\lim_{x \rightarrow 6} f(x) =$

b) Find and classify the discontinuities of f as removable, infinite or jump.

1. [10 pts.] Simplify the expression $\tan(\sin^{-1} x)$.

Let $y = \sin^{-1} x$. Then $\sin y = x$ and $y \in [-\pi/2, \pi/2]$.

Since $\sin^2 y + \cos^2 y = 1$, then $\cos y = \sqrt{1 - x^2}$.

Thus, $\tan(\sin^{-1} x) = \tan y = \frac{\sin y}{\cos y} = \frac{x}{\sqrt{1 - x^2}}$.

2. [10 pts.] Let $f(x) = \frac{2^x - 3}{2^x - 1}$. Find $f^{-1}(x)$.

(1) Let $y = \frac{2^x - 3}{2^x - 1}$.

(2) Solve for x in terms of y : $x = \log_2 \left(\frac{y - 3}{y - 1} \right)$.

(3) Interchange x and y : $y = \log_2 \left(\frac{x - 3}{x - 1} \right)$. Therefore, $f^{-1}(x) = \log_2 \left(\frac{x - 3}{x - 1} \right)$.

3. [10 pts.] Use the Intermediate Value Theorem to show that the equation $x^3 + x^2 - 2x - 1 = 0$ has a real root.

Let $f(x) = x^3 + x^2 - 2x - 1$.

The function f is continuous everywhere (it is a polynomial). Since $f(1) = -1 < 0$ and $f(2) = 7 > 0$, therefore, by the IVT $\exists c \in (1, 2)$ s.t. $f(c) = 0$. Hence, the given equation has a real root.

4. [10 pts.] Find a value for k , if any, such that $\lim_{x \rightarrow 1} \left(\frac{1}{x - 1} - \frac{k}{x^2 - 1} \right)$ exists and is finite.

$$\lim_{x \rightarrow 1} \left(\frac{1}{x - 1} - \frac{k}{x^2 - 1} \right) = \lim_{x \rightarrow 1} \left(\frac{x + 1 - k}{x^2 - 1} \right).$$

For this limit to exist it is necessary that $\lim_{x \rightarrow 1} x + 1 - k = 0$, i.e. $k = 2$.

When $k = 2$, $\lim_{x \rightarrow 1} \frac{x + 1 - 2}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{x - 1}{(x - 1)(x + 1)} = 1/2$.

5. [3+7=10 pts.] Let $f(x) = \frac{|x|}{x^2 - x}$.

(a) Find all points of discontinuity of f , if any.

f is discontinuous at $x = 0, 1$.

(b) Classify each discontinuity as removable, jump, or infinite (justify your answers).

$$\lim_{x \rightarrow 0^\pm} f(x) = \lim_{x \rightarrow 0^\pm} \frac{|x|}{x^2 - x} = \lim_{x \rightarrow 0^\pm} \frac{\pm x}{x(x - 1)} = \lim_{x \rightarrow 0^\pm} \frac{\pm 1}{(x - 1)} = \mp 1$$

Thus, f has a jump discontinuity at $x = 0$.

$$\lim_{x \rightarrow 1^\pm} f(x) = \lim_{x \rightarrow 1^\pm} \frac{|x|}{x^2 - x} = \pm \infty$$

Thus, f has an infinite discontinuity at $x = 1$.

6. [8+7 = 15 pts.] Evaluate each of the following limits, if it exists.

(a) $\lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x + 2} - 2} = \lim_{x \rightarrow 2} \frac{(x^2 - 4)(\sqrt{x + 2} + 2)}{(\sqrt{x + 2} - 2)(\sqrt{x + 2} + 2)}$

$$= \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)(\sqrt{x + 2} + 2)}{x - 2} = \lim_{x \rightarrow 2} (x + 2)(\sqrt{x + 2} + 2) = 16.$$

(b) $\lim_{x \rightarrow 0} x^2 e^{\sin(\frac{\pi}{x})}$.

Since $-1 \leq \sin\left(\frac{\pi}{x}\right) \leq 1$, it follows that $e^{-1} \leq e^{\sin(\frac{\pi}{x})} \leq e$. Hence, $x^2 e^{-1} \leq x^2 e^{\sin(\frac{\pi}{x})} \leq x^2 e$.

Since $\lim_{x \rightarrow 0} x^2 e^{-1} = \lim_{x \rightarrow 0} x^2 e = 0$, it follows from the Squeeze Theorem that $\lim_{x \rightarrow 0} x^2 e^{\sin(\frac{\pi}{x})} = 0$.

7. [7+8 = 15 pts.] Let $f(x) = \frac{\sqrt{4x^2 + 10}}{x - 5}$.

(a) State the domain of f , and find the vertical asymptotes of the graph of f , if any.

$D_f = (-\infty, 5) \cup (5, \infty)$.

$\lim_{x \rightarrow 5^\pm} f(x) = \lim_{x \rightarrow 5^\pm} \frac{\sqrt{4x^2 + 10}}{x - 5} = \pm\infty$.

Therefore, the line $x = 5$ is a vertical asymptote of the graph of f .

(b) Find the horizontal asymptotes of the graph of f , if any.

$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{\sqrt{4x^2 + 10}}{x - 5} = \lim_{x \rightarrow \pm\infty} \frac{\sqrt{x^2(4 + 10/x^2)}}{x(1 - 5/x)} = \lim_{x \rightarrow \pm\infty} \frac{\pm x \sqrt{4 + 10/x^2}}{x(1 - 5/x)} = \pm 2$.

Therefore, the lines $y = -2$ and $y = 2$ are horizontal asymptotes of the graph of f .

8. [10+10 = 20 pts.] Use the graph of $y = f(x)$ shown below to answer the followings:

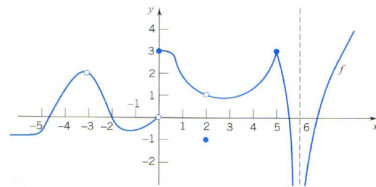


Figure 1: The graph of $y = f(x)$.

a) Find each of the following limits, if it exists:

i) $\lim_{x \rightarrow -3} f(x) = 2$

ii) $\lim_{x \rightarrow 0^-} f(x) = 0$

iii) $\lim_{x \rightarrow 0^+} f(x) = 3$

iv) $\lim_{x \rightarrow 2} f(x) = 1$

v) $\lim_{x \rightarrow 6} f(x) = -\infty$

b) Find and classify the discontinuities of f as removable, infinite or jump.

f is discontinuous at $x = -3, 0, 2, 6$.

f has a removable discontinuity at $x = -3$

f has a jump discontinuity at $x = 0$

f has a removable discontinuity at $x = 2$

f has an infinite discontinuity at $x = 6$.